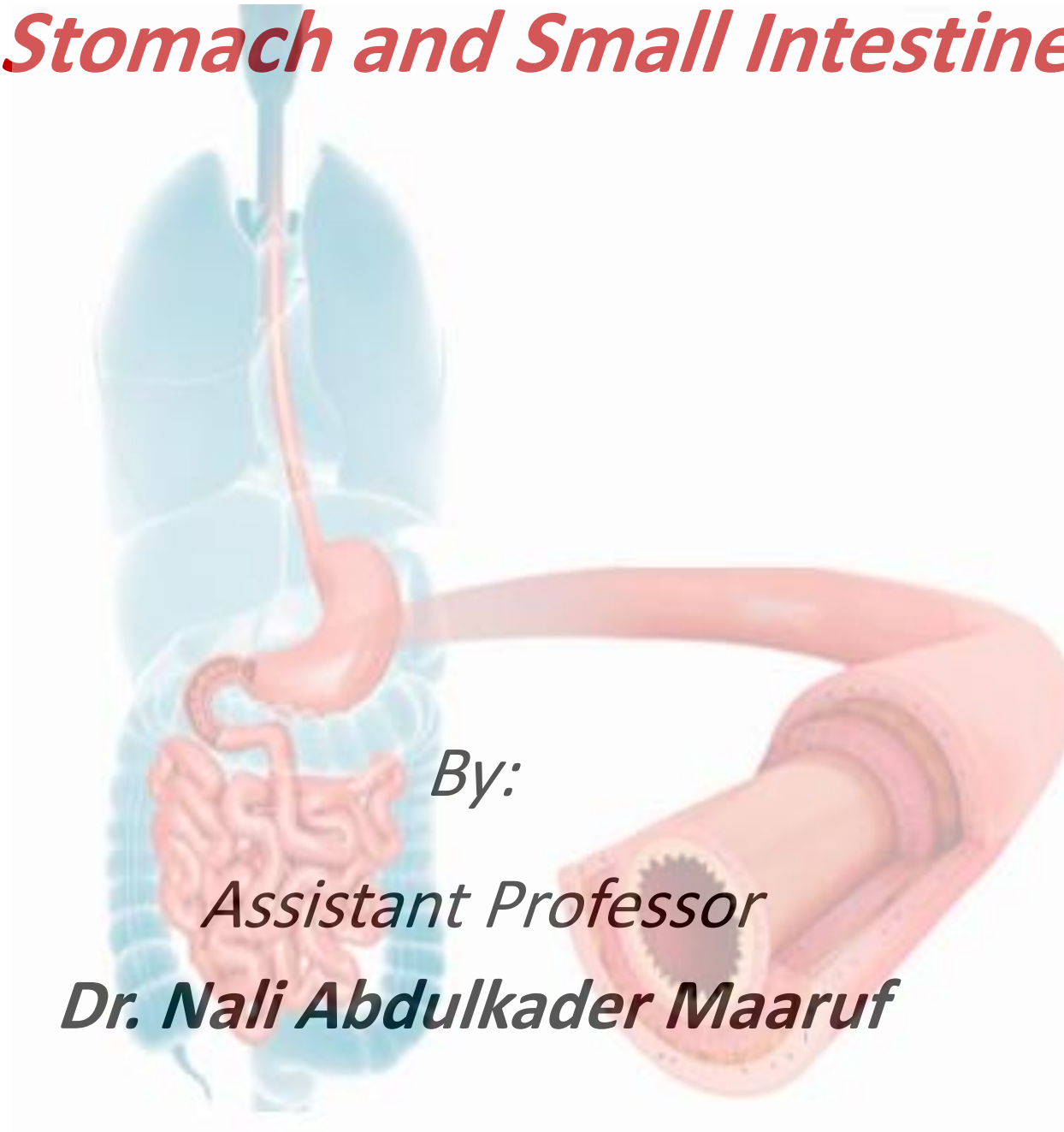


Stomach and Small Intestine



Objectives

At the end of the lecture, students should:

- ☐ List parts of stomach
- ☐ Describe the surfaces, openings and curvatures of the stomach
- ☐ Relations, blood supply and lymphatic drainage of the stomach
- ☐ List the different parts of small intestine.
- ☐ Describe the anatomy of duodenum, jejunum & ileum regarding: the shape, length, site of beginning & termination, peritoneal covering, arterial supply & lymphatic drainage.
- ☐ Differentiate between each part of duodenum regarding the length, level & relations.
- ☐ Differentiate between the jejunum & ileum regarding the characteristic anatomical features of each of them.

STOMACH

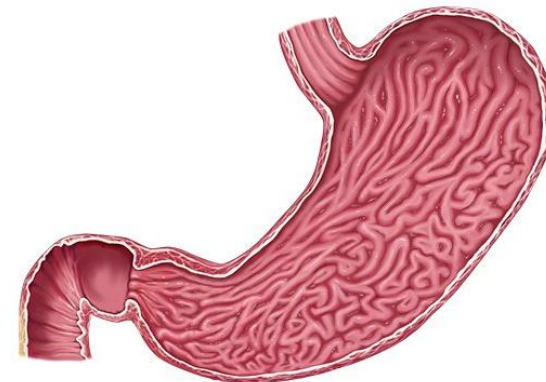
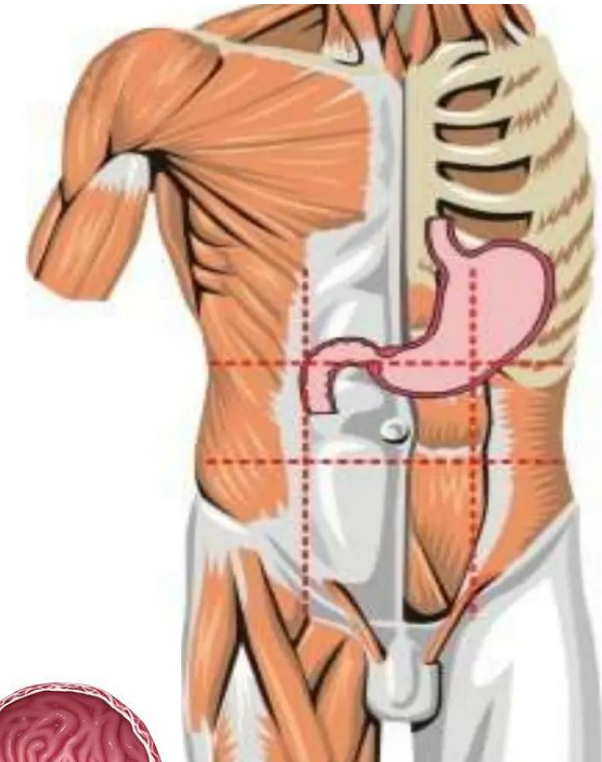
❑ The stomach is the dilated part of the alimentary canal that stores food and mixes it with gastric secretions to form a semifluid called chyme.

○ **Location:** it occupies the left hypochondriac region and the epigastric and umbilical regions, and it extends from the level of T10 to L3.

○ **Shape and size:** An empty stomach has a J-shape and sits obliquely, horizontally in obese individuals called (a teapot-shaped stomach)

It is about 10 inches long, with a capacity of approximately 1000 ml during puberty and 1500–2000 ml in adulthood.

○ **Interior of the stomach:** longitudinal ridges called longitudinal gastric folds (rugae), which are most marked toward the pyloric part along the greater curvature



External features

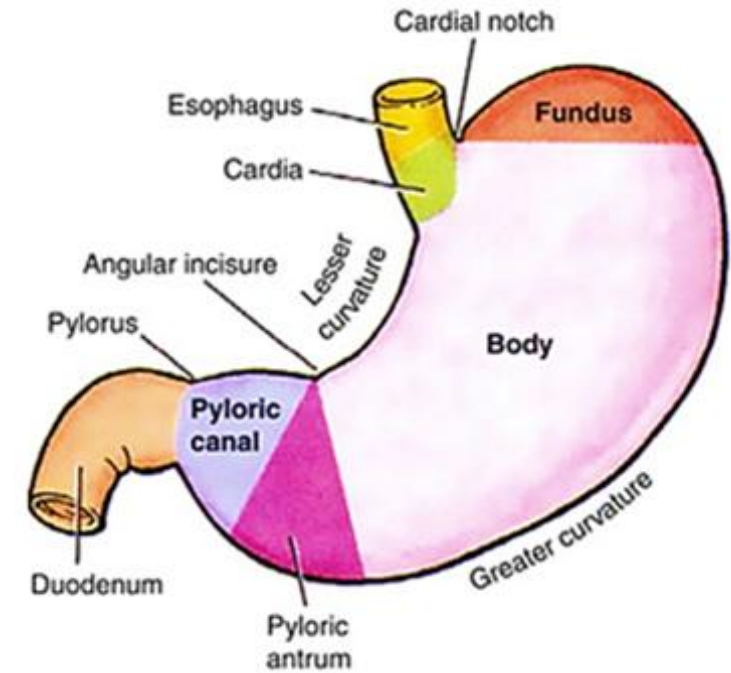
❑ The stomach has two openings, two curvatures, two surfaces, and four parts.

❑ **Two openings are cardiac and pyloric.**

1) **Cardiac opening:** It is the cranial opening that is continuous with the oesophagus, and it is located behind the left 7th costal cartilage, 1 inch away from the sternum, at the level of T11.

2) **Pyloric opening:** Is the caudal opening, which is continuous with the duodenum and is located at the level of L1 (at the transpyloric plane). It is about 1.5 cm to the right of the median plane.

Pylorus' is derived from a Greek word meaning 'gatekeeper', as the anatomical sphincter controls this opening called the pyloric sphincter.



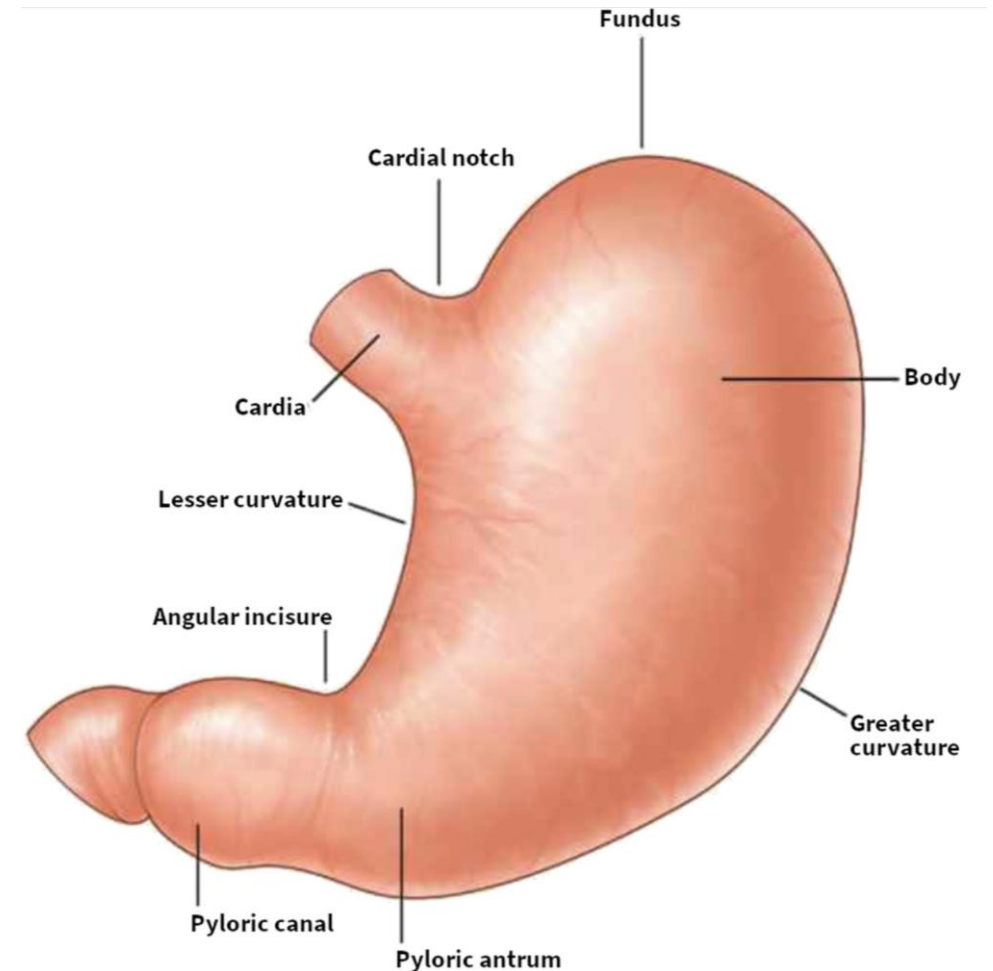
❑ Two curvatures are greater and lesser.

1. Greater Curvature:

- Is convex, forms the left border of the stomach, and is five times longer than the lesser curvature.
- It provides attachment to the greater omentum.
- At the upper end there is a cardiac notch that separates it from the oesophagus.

2. Lesser Curvature:

- Is concave, forming the right border of the stomach
- It provides attachment to the lesser omentum.
- At the lower end there is an incisura angularis.

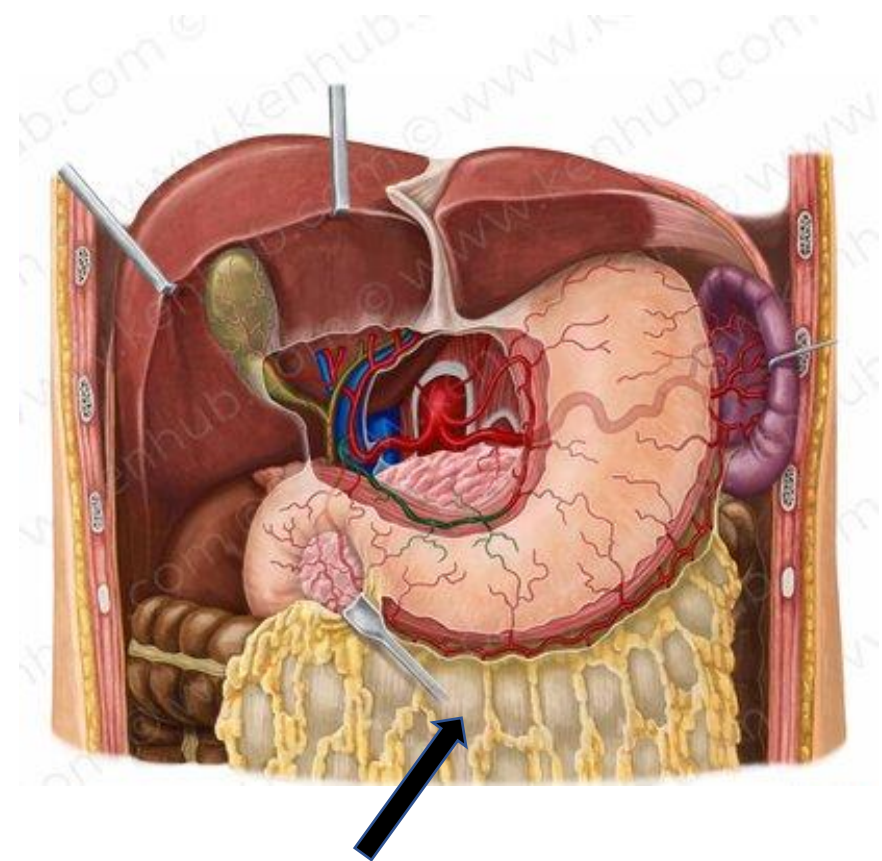


Greater Omentum (L. Omentum = Apron)

The greater omentum is a large, thick fold of peritoneum extends from the greater curvature of the stomach and one inch of the duodenum to the transverse colon and transverse mesocolon.

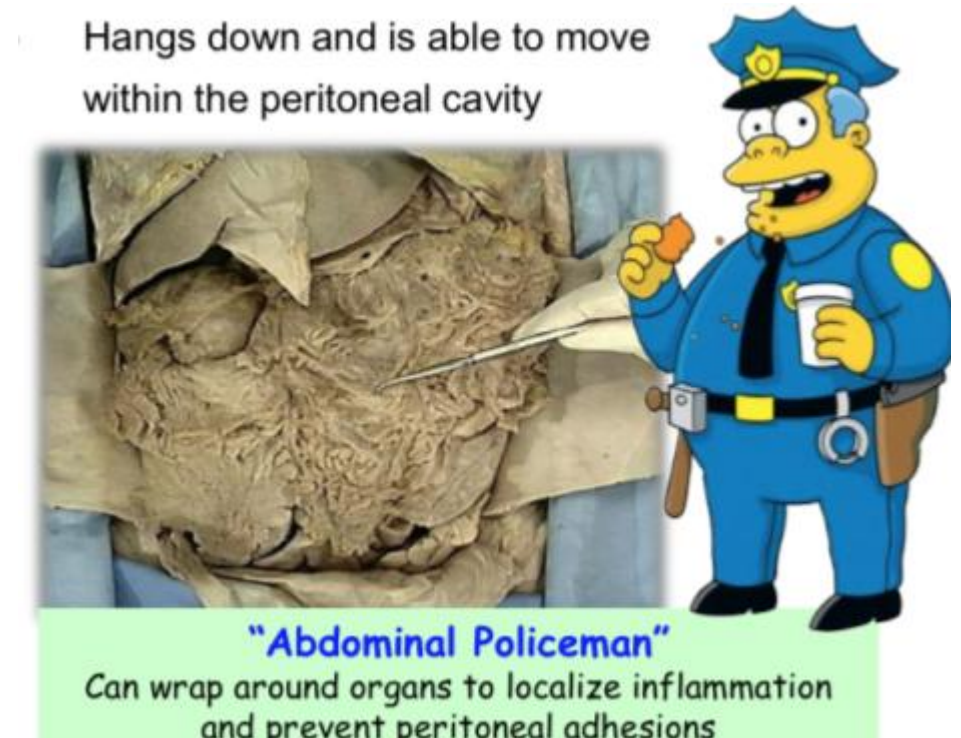
Contents

- Adipose tissue (fat) in varying amounts dependent on the individual's nutritional status.
- Aggregations of macrophages that form dense patches known as milky spots.
- Right and left gastroepiploic vessels.



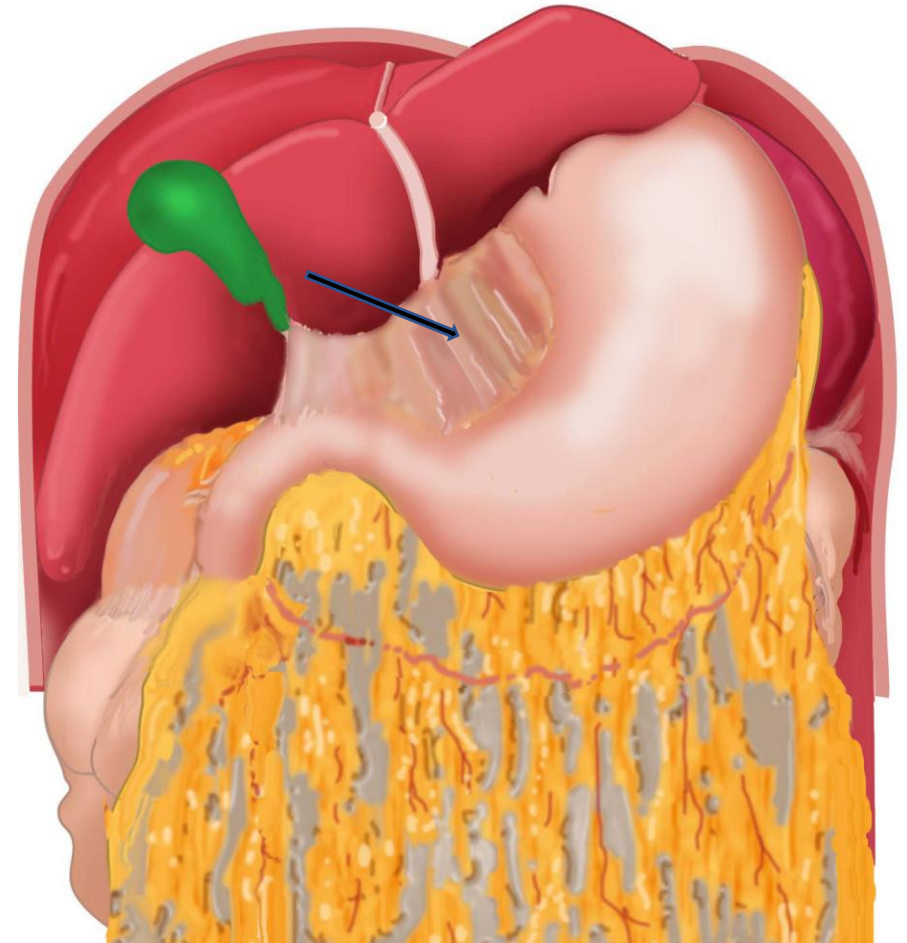
CLINICAL ANATOMY

- ❑ The greater omentum protects the peritoneal cavity from infection due to the presence of a large number of macrophages.
 - ❑ It limits the spread of infection by moving to the site of infection and sealing it off from surrounding areas.
 - ❑ For this reason, the greater omentum is often referred to as the "policeman of the abdomen."
- Additionally, it serves as a storehouse for adipose tissue and is sometimes used as grafting material by surgeons.



Lesser Omentum

- ❑ The lesser omentum is a double-layered fold of peritoneum that extends between the lesser curvature of the stomach and the inferior surface of the liver.
- ❑ Inferiorly, it attaches to the right side of the abdominal esophagus, the lesser curvature of the stomach, and the first 2 cm of the duodenum.
- ❑ Superiorly, it connects in an inverted L-shaped manner to the margins of the fissures for the ligamentum venosum and the porta hepatis.



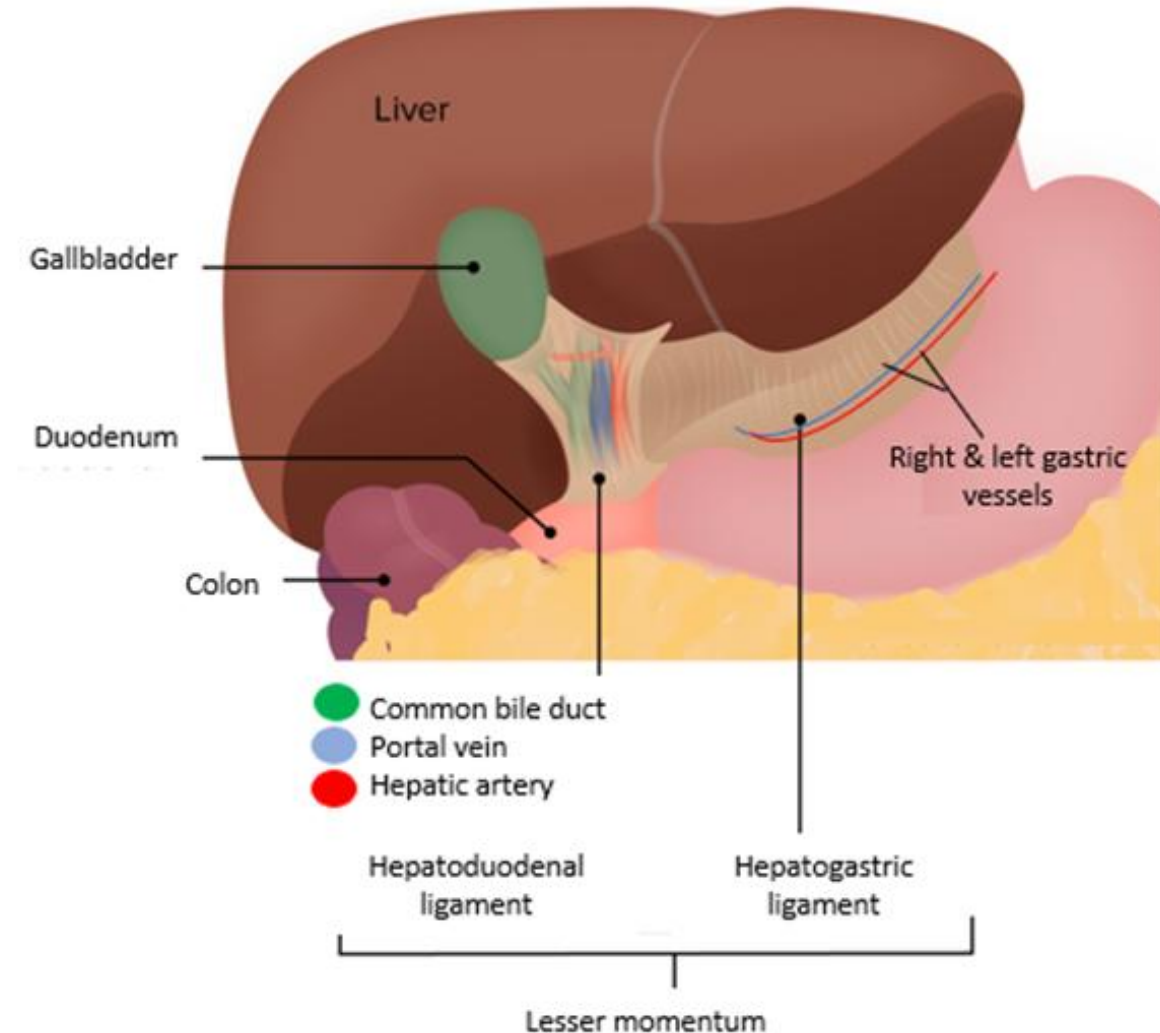
Contents of the lesser Omentum

□ In the free margin (hepatoduodenal ligament) superficial → deep:

- Bile duct (right side, anterior)
- Hepatic artery proper (left side, anterior)
- Portal vein (posterior, deepest)

□ In the hepatogastric ligament (superficial → deep):

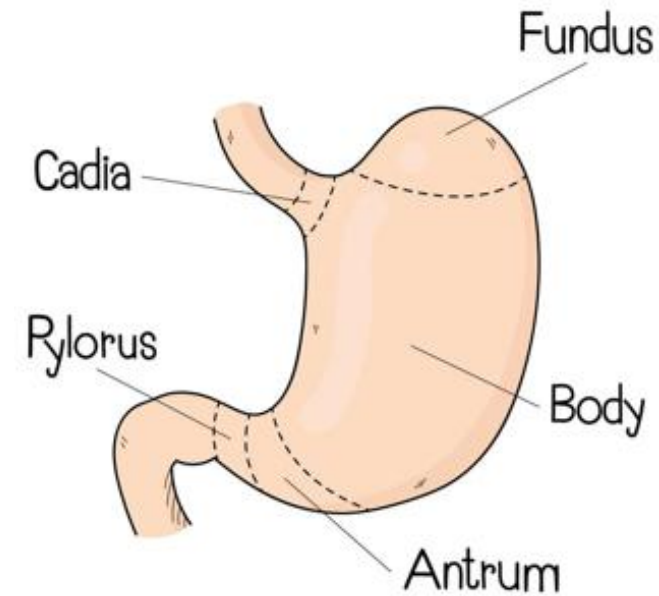
- Right gastric vessels
- Left gastric vessels
- Lymphatics & lymph nodes
- Branches of vagus nerves
- Sympathetic fibers from celiac plexus



❑ Four parts (Cardia, Fundus, Body and Pyloric):

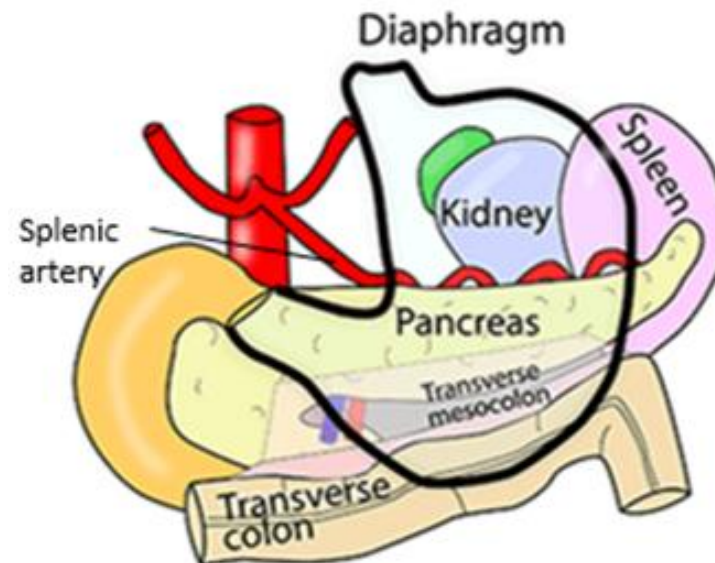
1. **Cardia** Located to the left of the midline at the level of the T11 vertebra (just below the xiphisternal joint).
2. **Fundus** The dome-shaped part is located beneath the left dome of the diaphragm, found at the level of the 5th left rib in the midclavicular line and often visible as a gas bubble on X-ray.
3. **Body:** The largest part of the stomach, extends from the fundus to the incisura angularis (the notch on the lesser curvature).
4. **Pyloric Part** divided into:
 - o **Pyloric antrum:** the wider proximal part
 - o **Pyloric canal:** the narrow terminal part that leads to the pylorus (sphincter).

Located to the right of the midline, at the level of the L1 vertebra (transpyloric plane).



❑ **Two surfaces are anterior (anterior superior) and posterior (posterior inferior).**

- **Anterior surface:** is related to the anterior abdominal wall, the left costal margin, the left pleura and lung, the diaphragm, and the left lobe of the liver
- **Posterior surface:** Is related to the lesser sac, the diaphragm, and the stomach bed (on which the stomach rests in the supine position), which is formed by the spleen, the left suprarenal gland, the upper part of the left kidney, the splenic artery, the pancreas, the transverse mesocolon, and the transverse colon



mnemonic : *Dr Steve Smith kills patients cruelly and mercilessly.*

- ✓ D- Diaphragm
- ✓ S- Suprarenal gland
- ✓ K – Kidney (Left)
- ✓ P – Pancreas (Body)
- ✓ C- Colic flexure (Left)
- ✓ M- Mesocolon (transverse)

Arterial supply of the stomach

The arteries are branches of the celiac artery

- The left gastric artery arises from the celiac artery: It supplies the lower third of the esophagus and the upper right part of the stomach.

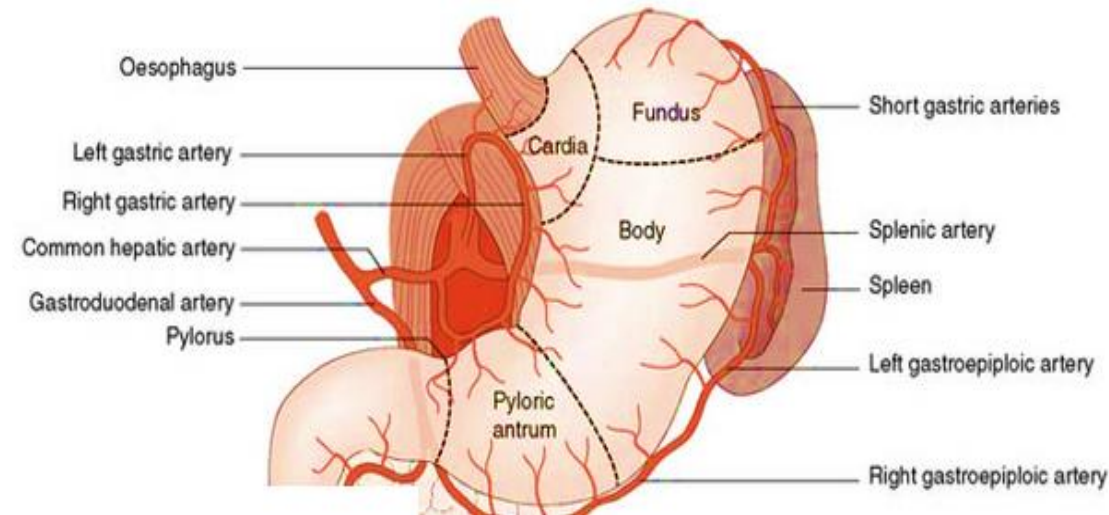
- The right gastric artery arises from the hepatic artery.

It supplies the lower right part of the stomach.

- The short gastric arteries arise from the splenic artery to supply the fundus.

- The left gastroepiploic artery arises from the splenic artery, the upper part of the greater curvature.

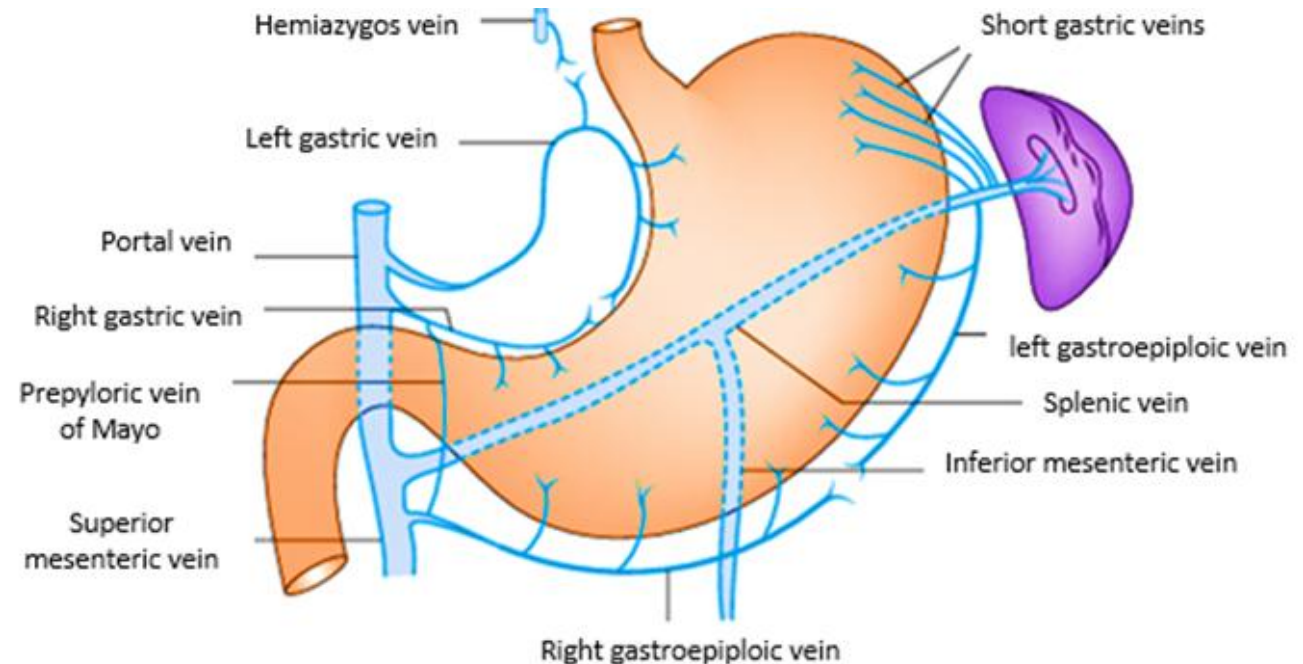
- The right gastroepiploic artery arises from the gastroduodenal branch of the hepatic artery and supplies the lower part of the greater curvature.



Venous drainage

❑ The veins drain into the portal circulation

- The left and right gastric veins drain directly into the **portal vein**.
- The short gastric veins and the left gastroepiploic veins join the **splenic vein**.
- The right gastroepiploic vein joins the **superior mesenteric vein**.



CLINICAL ANATOMY

- ❑ The anastomoses of the arteries that supply the stomach provide adequate collateral circulation, allowing one or more arteries to be ligated during total or partial gastrectomy without seriously affecting the blood supply to the remaining part of the stomach.
- ❑ When removing the pyloric antrum, the greater omentum is incised parallel and inferior to the right gastro-omental artery, necessitating ligation of all the omental branches of this artery

Lymph Drainage

- Lymph vessels follow the arteries to the left and right gastric nodes, the left and right gastroepiploic nodes, and the short gastric nodes. Ultimately, all lymph from the stomach is directed to the celiac nodes, which are located around the root of the celiac artery on the posterior abdominal wall.

Nerve supply

- The stomach supplied by sympathetic and parasympathetic fibers
 - Sympathetic fibers from T6–T9 spinal segments (via greater splanchnic nerves, celiac plexus)
 - Parasympathetic from the right and left vagus.

Anterior vagal trunk, which is formed by the left vagus nerve supply the anterior surface of the stomach.

Posterior vagal trunk, which is formed by the right vagus nerve, supply mainly the posterior surface of the stomach.

CLINICAL ANATOMY

- ❑ **Vagotomy:** The vagus nerves play a significant role in controlling the secretion of acid by the parietal cells of the stomach. Since excessive acid secretion is a primary cause of peptic ulcers, a surgical procedure known as vagotomy involves sectioning the vagal trunks as they enter the abdomen to reduce acid production.
- ❑ **Gastric Pain:** This type of pain is felt in the epigastric region because the stomach is innervated by the T6 to T9 segments of the spinal cord, which also supply the upper part of the abdominal wall.
- ❑ **Gastric Ulcer:** Commonly seen in individuals who experience stress and consume spicy foods, gastric ulcers typically occur along the lesser curvature of the stomach due to several factors: The epithelium is thin, Blood supply is reduced, with fewer anastomoses., The presence of the gastric canal that receives chyme, Shorter length of the affected area.

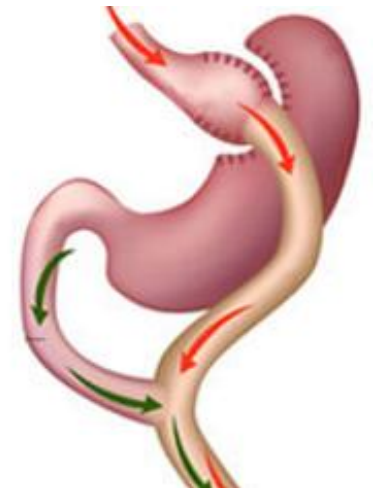
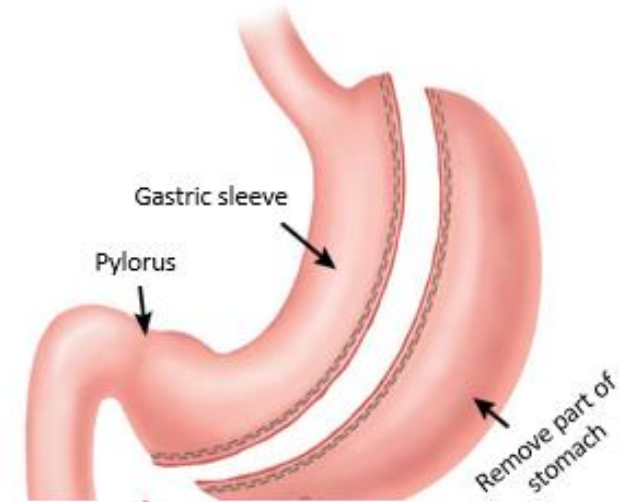
CLINICAL ANATOMY

□ Gastric ulcer

- A posterior gastric ulcer can erode through the stomach wall into the pancreas, resulting in referred pain to the back. In such cases, erosion of the splenic artery can lead to severe haemorrhage
- Erosion of the anterior wall may allow gastric contents to escape into the greater sac, potentially affecting the liver, and may descend to the right iliac fossa, making the diagnosis of appendicitis difficult.
- **Gastric Carcinoma (Gastric Cancer):** This type of cancer often arises in the pyloric antrum along the greater curvature of the stomach. It commonly spreads via lymphatic vessels to the left supraclavicular lymph nodes. An enlarged and palpable left supraclavicular node, known as Virchow's node, may be the first indication of gastric cancer (referred to as Troisier's sign). The cancer cells reach the left supraclavicular lymph node through the thoracic duct.

CLINICAL ANATOMY

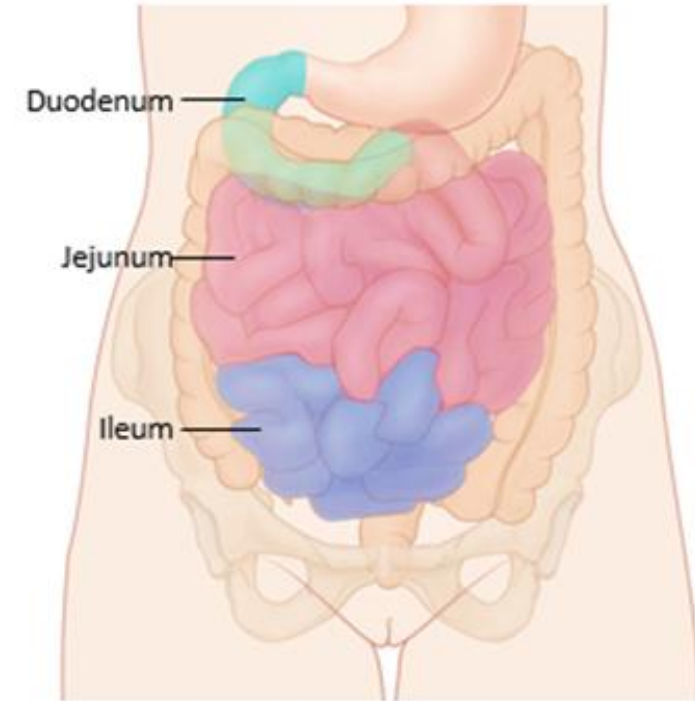
- **Hiatus Hernia:** This condition involves the stomach protruding through the diaphragm, which can cause reflux and dysphagia.
- **Sleeve Gastrectomy:** This weight-loss surgery involves the removal of a large portion of the stomach, particularly from the greater curvature. The remaining stomach is shaped like a narrow tube or sleeve along the lesser curvature. This smaller stomach helps individuals feel full quickly, thereby reducing food intake
- **Gastric Bypass:** Makes small stomach pouch + bypasses part of small intestine



The small intestine

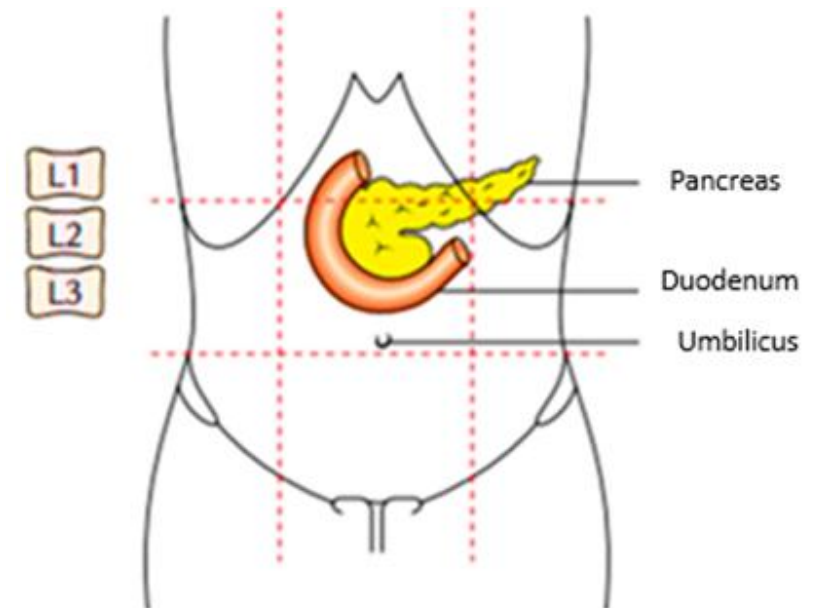
- ❑ Is the longest part of the digestive
- ❑ It is about 20 feet (6 meters) long in adults
- ❑ It performs 90% of digestion and nutrient absorption, leaving only 10% for the stomach and large intestine.
- ❑ The small intestine is suspended by the mesentery and divided into three parts the duodenum, the jejunum, and the ileum.
- ❑ Without the small intestine's functions, survival would be impossible

It is a digestive superpower



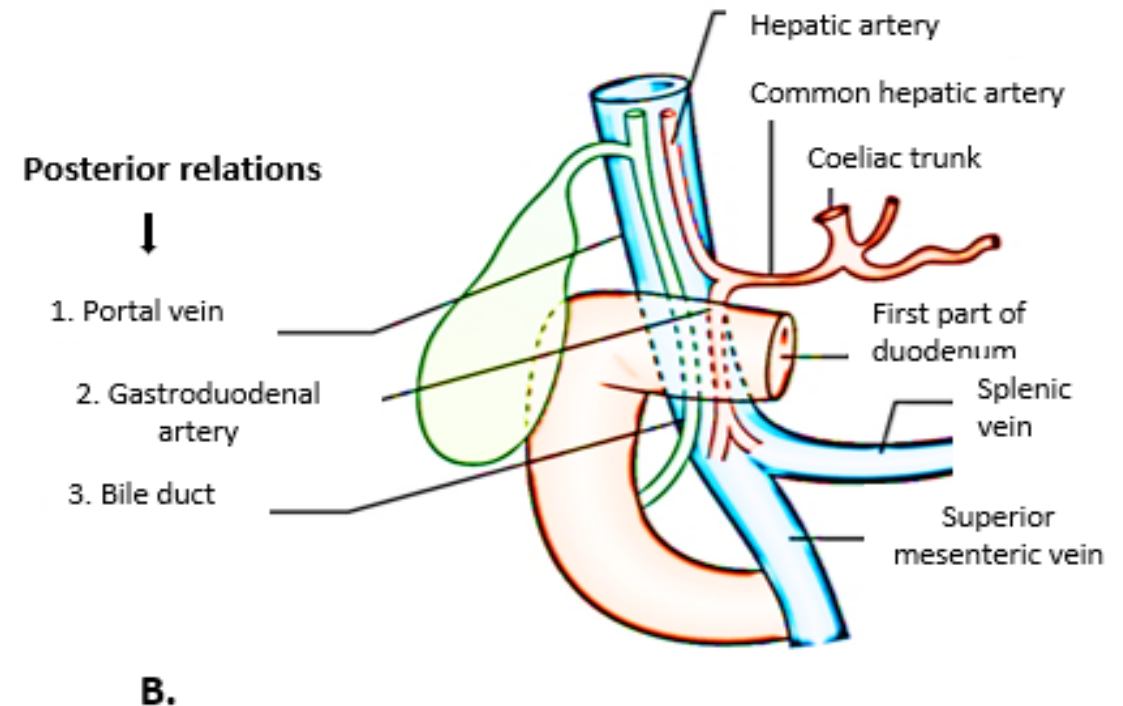
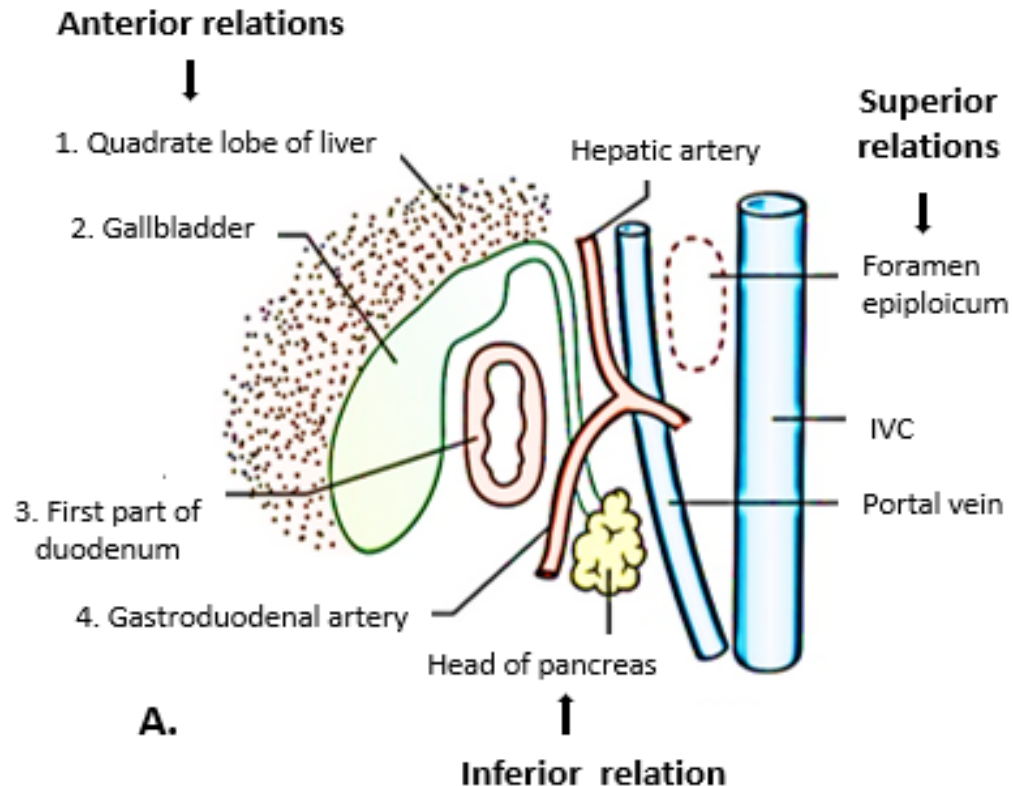
DUODENUM

- ❑ The duodenum is the first, shortest, widest, and most fixed part of the small intestine.
- ❑ The term “duodenum” is the Latin corruption of the Greek word “do-deka-daktulos,” meaning 12 fingers. It is C-shaped and about 10 inches (25 cm) long, extending from the pylorus to the duodenojejunal flexure and curving around the head of the pancreas.
- ❑ It is located in the abdominal cavity in the umbilical region, opposite the L1, L2, and L3 vertebrae.
- ❑ The duodenum is retroperitoneal except for the proximal 2.5 cm, which is suspended above by the lesser omentum and below by the greater omentum.
- ❑ For the purpose of study, the duodenum is divided into four parts.



First part (Superior Part)

- It is about 2 inches long; the first inch is intraperitoneal, and the second inch is retroperitoneal.
- Locates at the level L1
- Only part supplied by end arteries and only part that is devoid of circular mucus folds



CLINICAL CORRELATION

- ❑ The first part of the duodenum is a common site for duodenal ulcers, as it receives acidic chyme from the stomach.
- ❑ An ulcer on the anterior wall of the first inch of the duodenum may perforate into the upper part of the greater sac, above the transverse colon, and cause an ulcer of the liver and gallbladder, with fluid potentially escaping into the right lateral paracolic gutter and down to the right iliac fossa. Differentiating between a perforated duodenal ulcer and a perforated appendix may be difficult.
- ❑ An ulcer on the posterior wall of the first part of the duodenum may penetrate the wall and erode the relatively large gastroduodenal artery, causing severe hemorrhage, (65%) duodenal ulcers occur in the posterior wall of the first part

Second part (Descending part)

- ❑ The length is approximately 2 inches, and it is located at the levels of L1 to L3.
- ❑ On the posteromedial wall, it receives the common bile duct and main pancreatic duct through the major duodenal papilla, and occasionally the accessory pancreatic duct via the minor duodenal papilla.
- ❑ **Major duodenal papilla (greater duodenal papilla):** This opening allows for the passage of bile and pancreatic duct at the hepatopancreatic ampulla (also known as the ampulla of Vater)
- ❑ **Minor duodenal papilla (if present):** it is located about 2 cm superior to the major papilla, and it receives the accessory pancreatic duct (of Santorini).

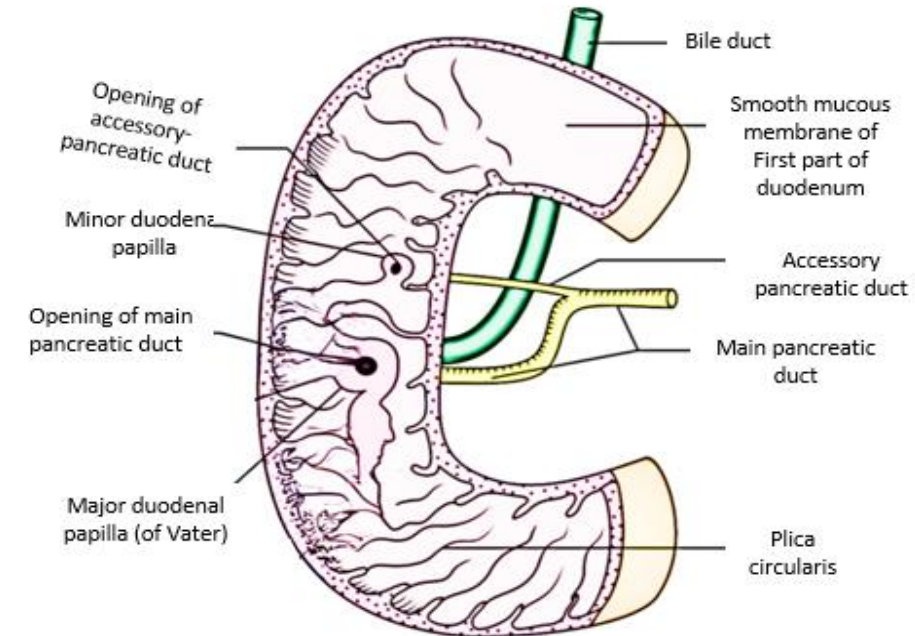


Fig. 59: Openings of the second part of the duodenum

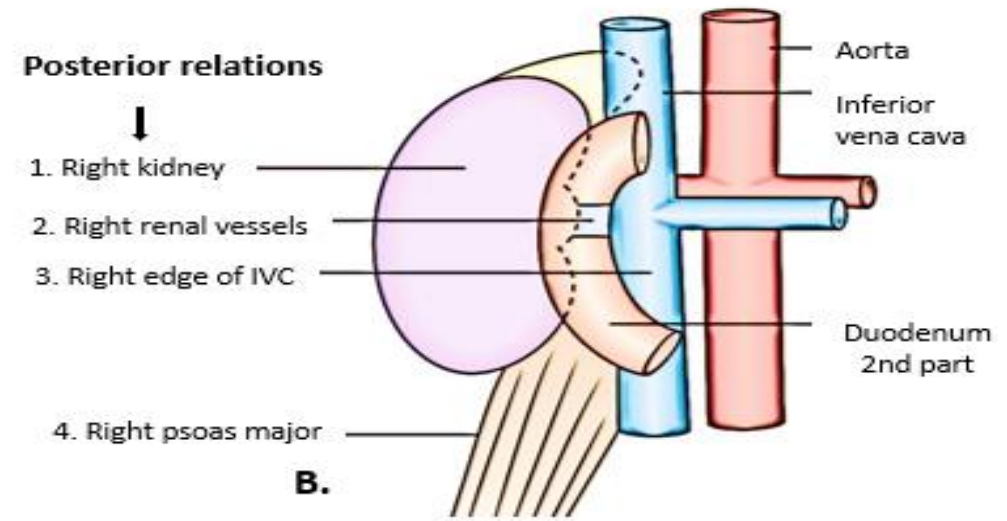
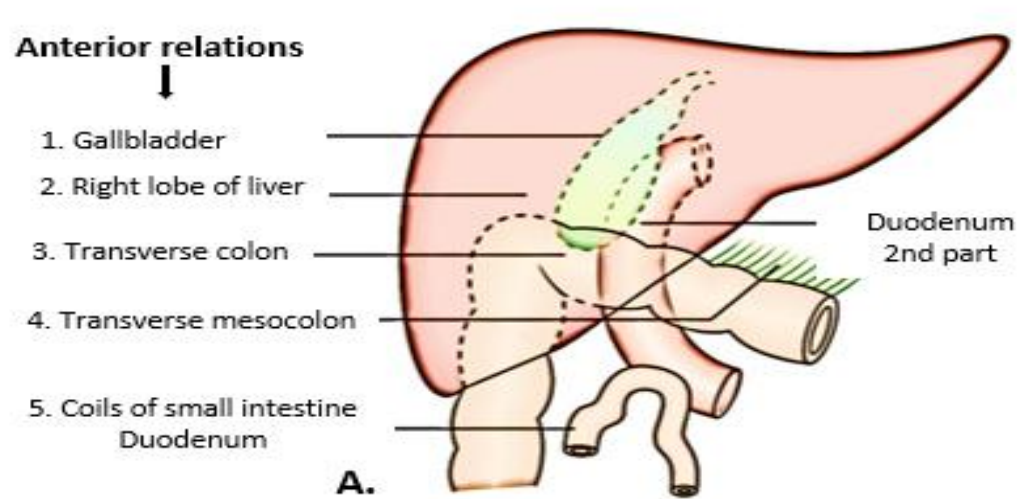


Fig. 60: Relation of the second part of the duodenum **A.** Anterior **B.** Posterior

CLINICAL CORRELATION.

Duodenal diverticula are congenital conditions that typically occur in the medial wall of the second part of the duodenum.

Third part of the duodenum

- It is about 4 inches long and runs horizontally to the left, across the lower part of the body of the L3 vertebra.

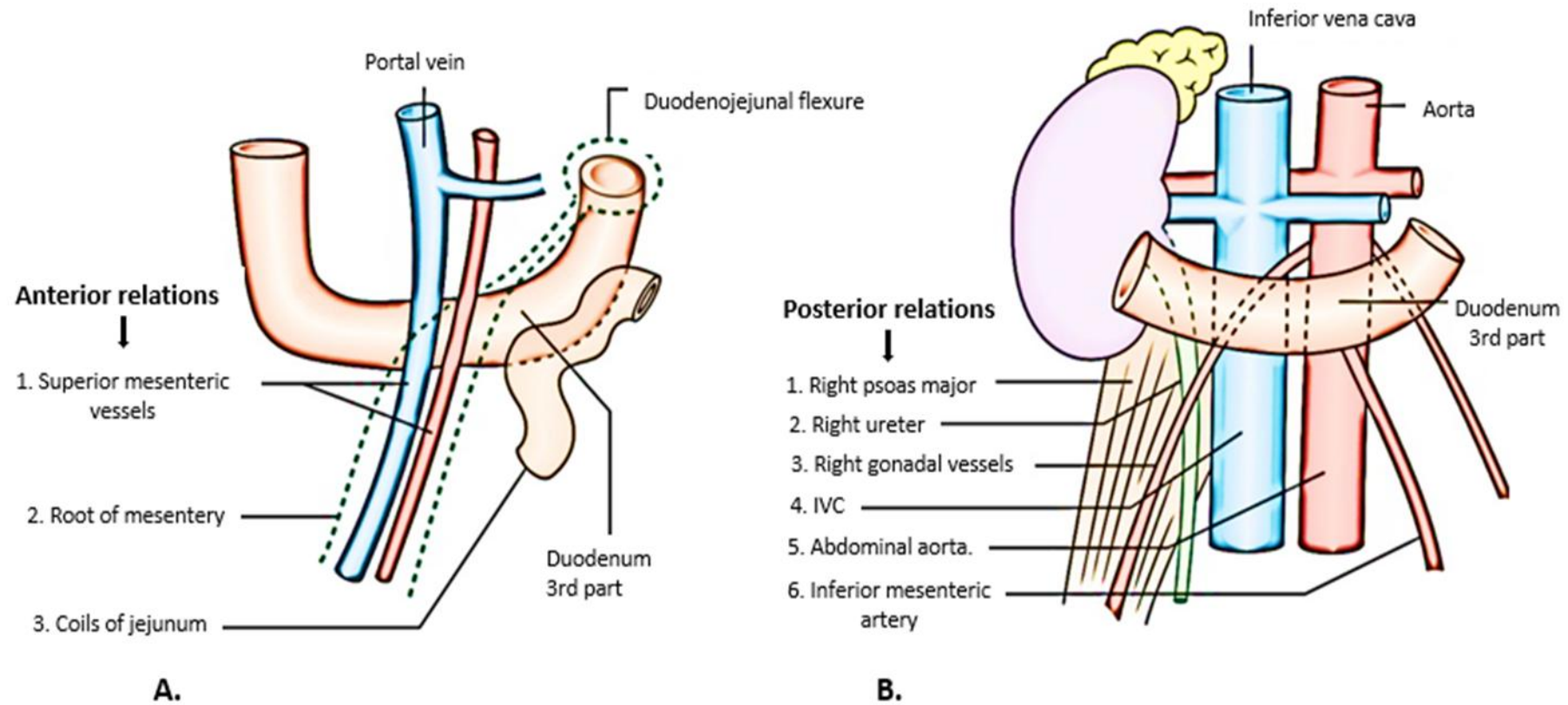


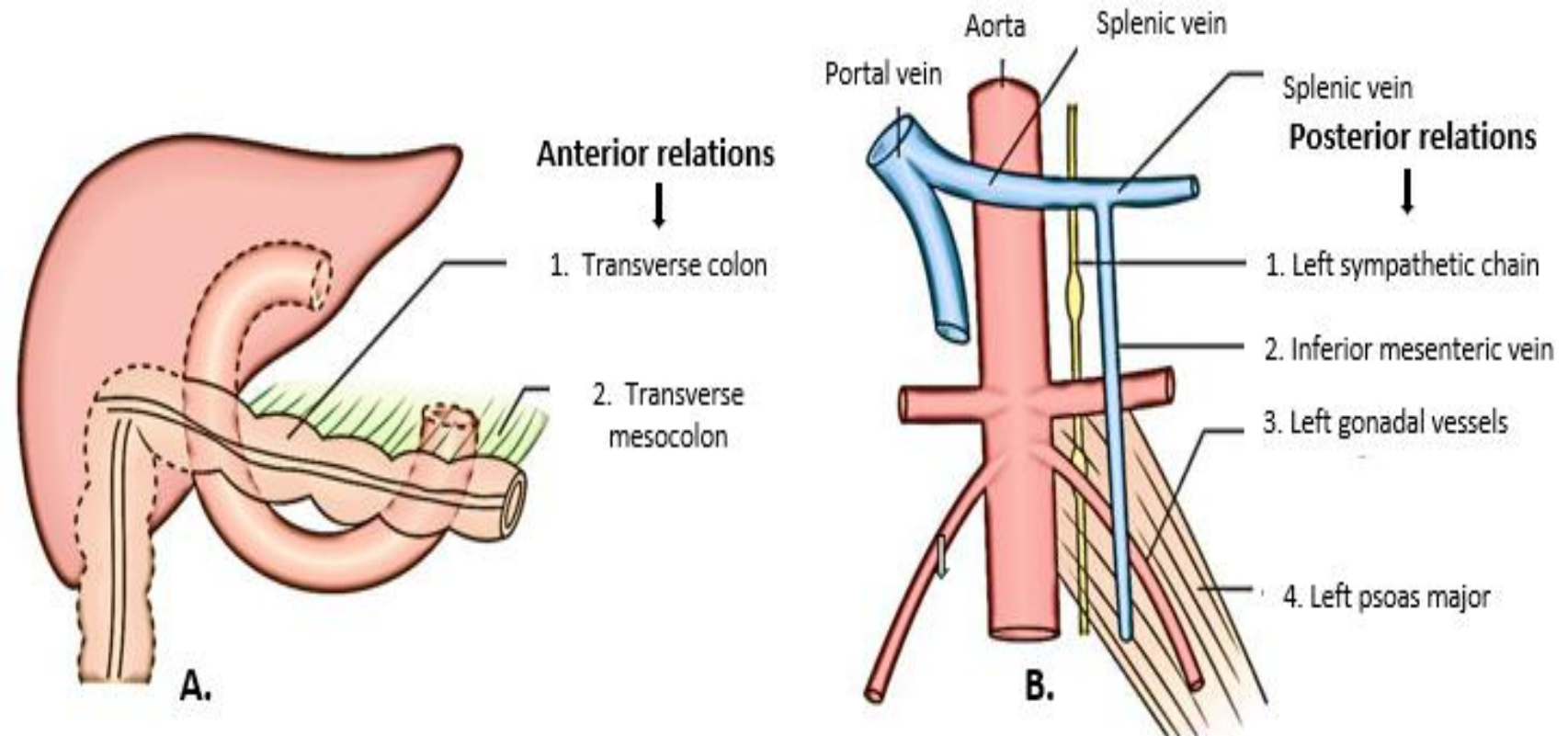
Fig. 61: Relation of the third part of the duodenum **A.** Anterior **B.** Posterior

CLINICAL CORRELATION

- Duodenal injuries: The second part of the duodenum is most protected from external injury because it lies in the paravertebral gutter—The third part of the duodenum is most vulnerable to external injury
- SMA syndrome is the compression of the third part of the duodenum (horizontal/inferior part) between the superior mesenteric artery (SMA) and the abdominal aorta, leading to partial or complete duodenal obstruction. The syndrome should be suspected in young patients with rapid weight loss who present with postprandial pain, nausea, and vomiting

Fourth part of the duodenum:

- It is about 1 inch long, extending from L3 to L2 to become continuous with the jejunum at the duodenojejunal flexure.

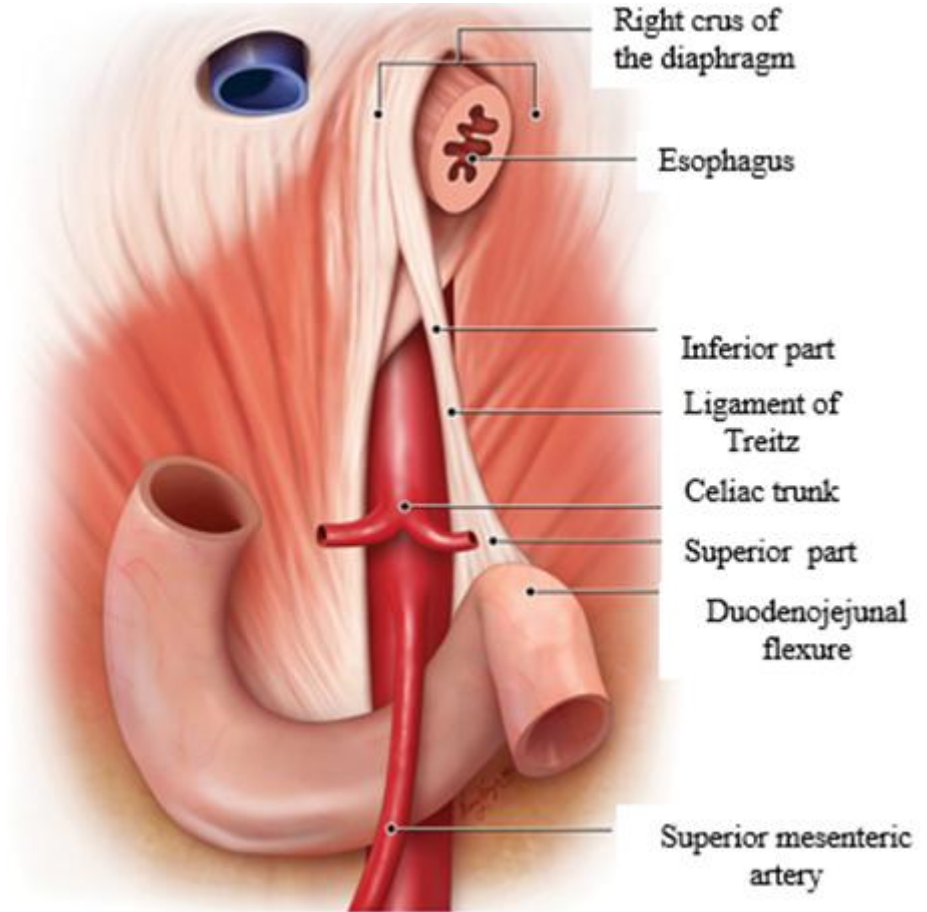


CLINICAL CORRELATION

- The relations to the duodenum of the gallbladder, the transverse colon, and the right kidney should be remembered. Cases have been reported in which a large gallstone ulcerated through the gallbladder wall into the duodenum.
- Operations on the colon and right kidney have resulted in damage to the duodenum.

Ligament of Treitz

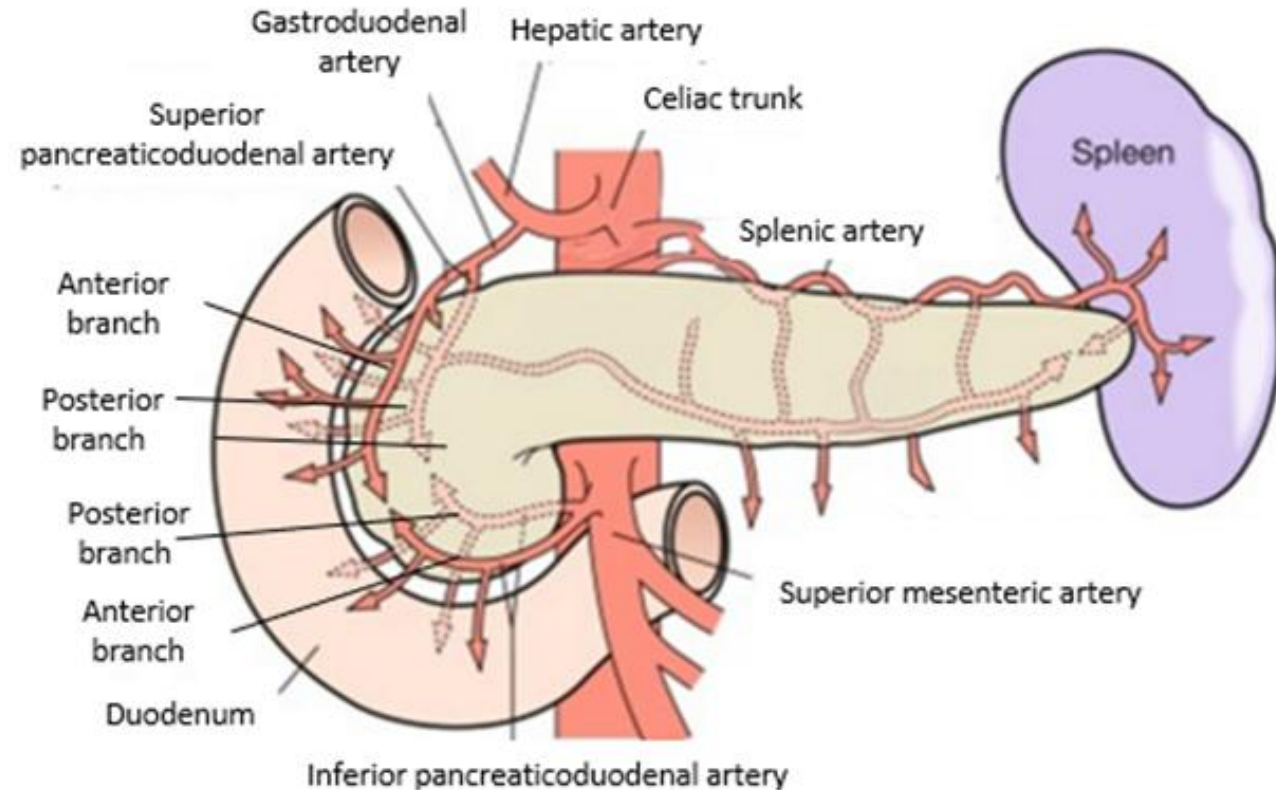
- ❑ It is a fibromuscular band that suspends the duodenojejunal flexure from the right crus of the diaphragm
- ❑ This ligament prevents the duodenojejunal flexure from being pulled down by the weight of the small intestine loops.
- ❑ Additionally, it serves as an important landmark for distinguishing between the upper gastrointestinal (GI) tract and the lower gastrointestinal (GIT)
- ❑ Occasionally, the ligament of Treitz may kink the duodenojejunal flexure, potentially leading to partial intestinal obstruction.



Arterial supply

❑ **Upper Half:** The upper portion of the duodenum is supplied by the superior pancreaticoduodenal artery, a branch of the gastroduodenal artery, which originates from the common hepatic artery of the celiac trunk.

❑ **Lower Half:** The lower portion receives blood supply from the inferior pancreaticoduodenal artery, a branch of the superior mesenteric artery.

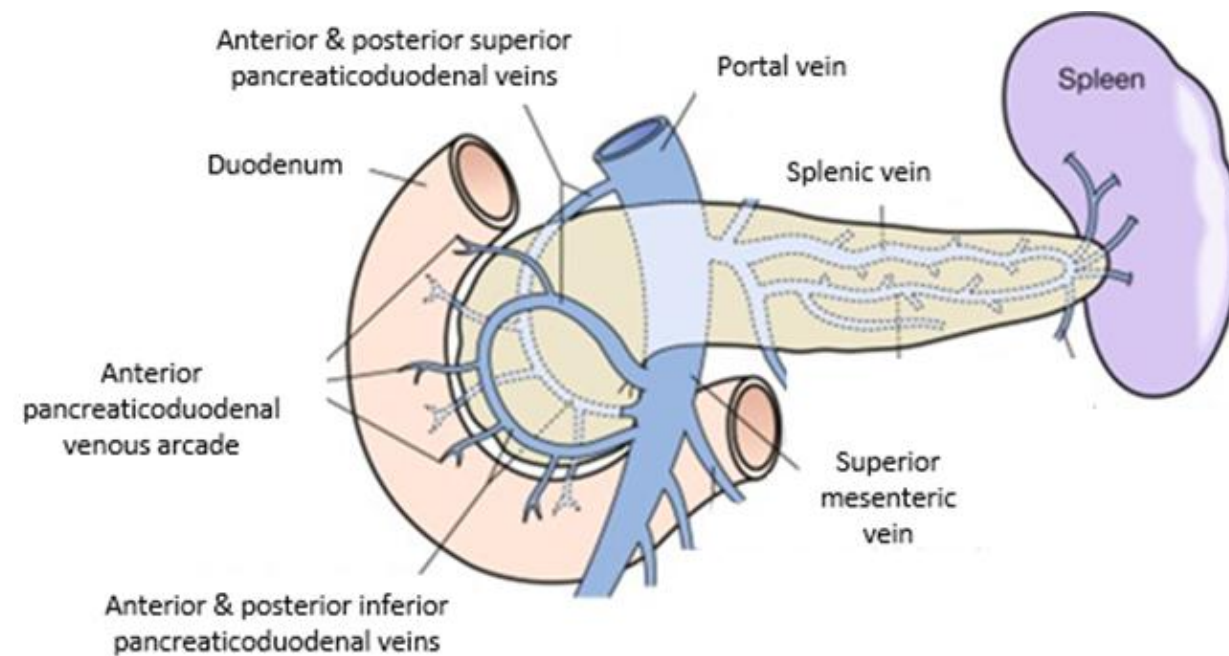


Venous Drainage

- ❑ Superior pancreaticoduodenal vein drains into the portal vein via the gastroduodenal or splenic veins.
- ❑ The inferior pancreaticoduodenal vein drains into the superior mesenteric vein.

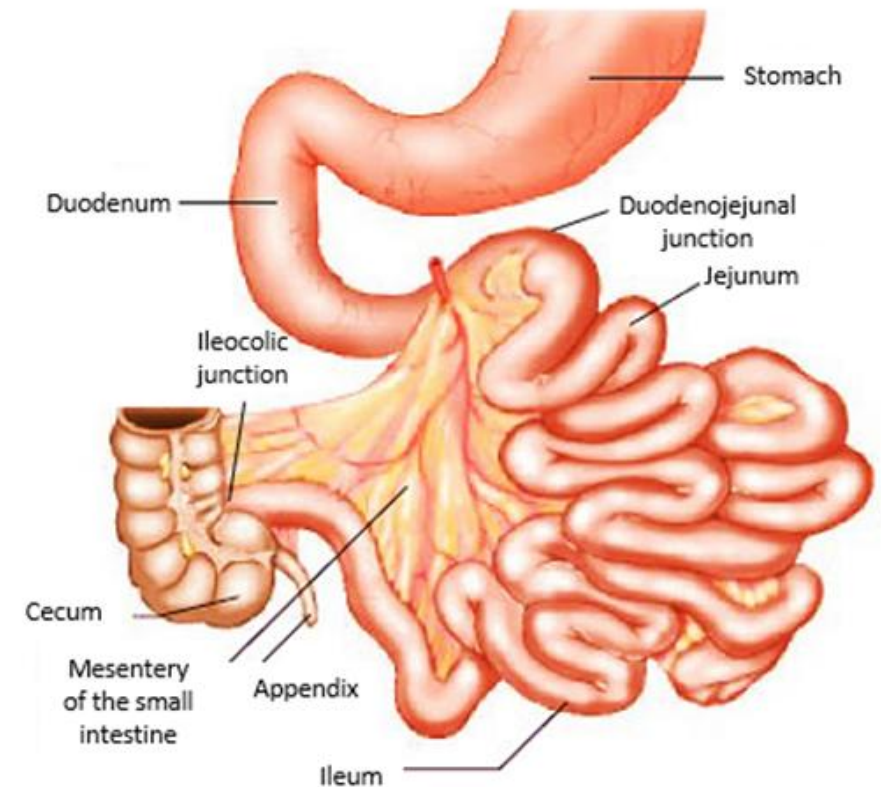
Lymph Drainage

- ❑ The upper duodenum drains upward through the pancreaticoduodenal nodes to the gastroduodenal nodes and then to the celiac nodes.
- ❑ the lower duodenum drains downward through pancreaticoduodenal nodes to the superior mesenteric nodes located at the origin of the superior mesenteric artery.



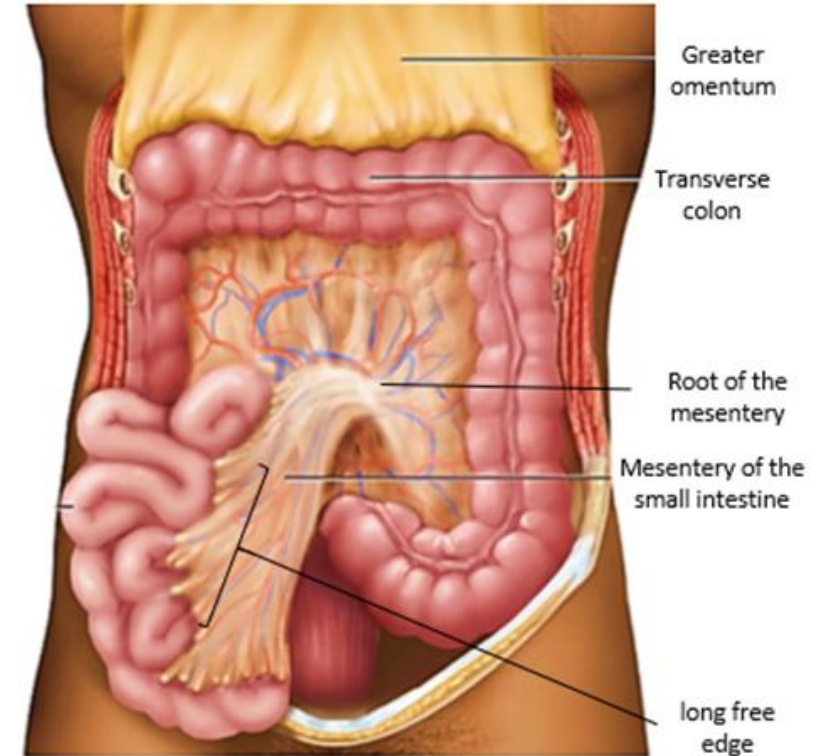
JEJUNUM AND ILEUM

- ❑ Sometimes called the small intestine proper, it extends from the duodenojejunal flexure to the ileocecal junction and is about 20 ft (6 m) long •
- ❑ Its upper two-fifths form the jejunum, and its lower three-fifths form the ileum suspended by mesentery of the small intestine



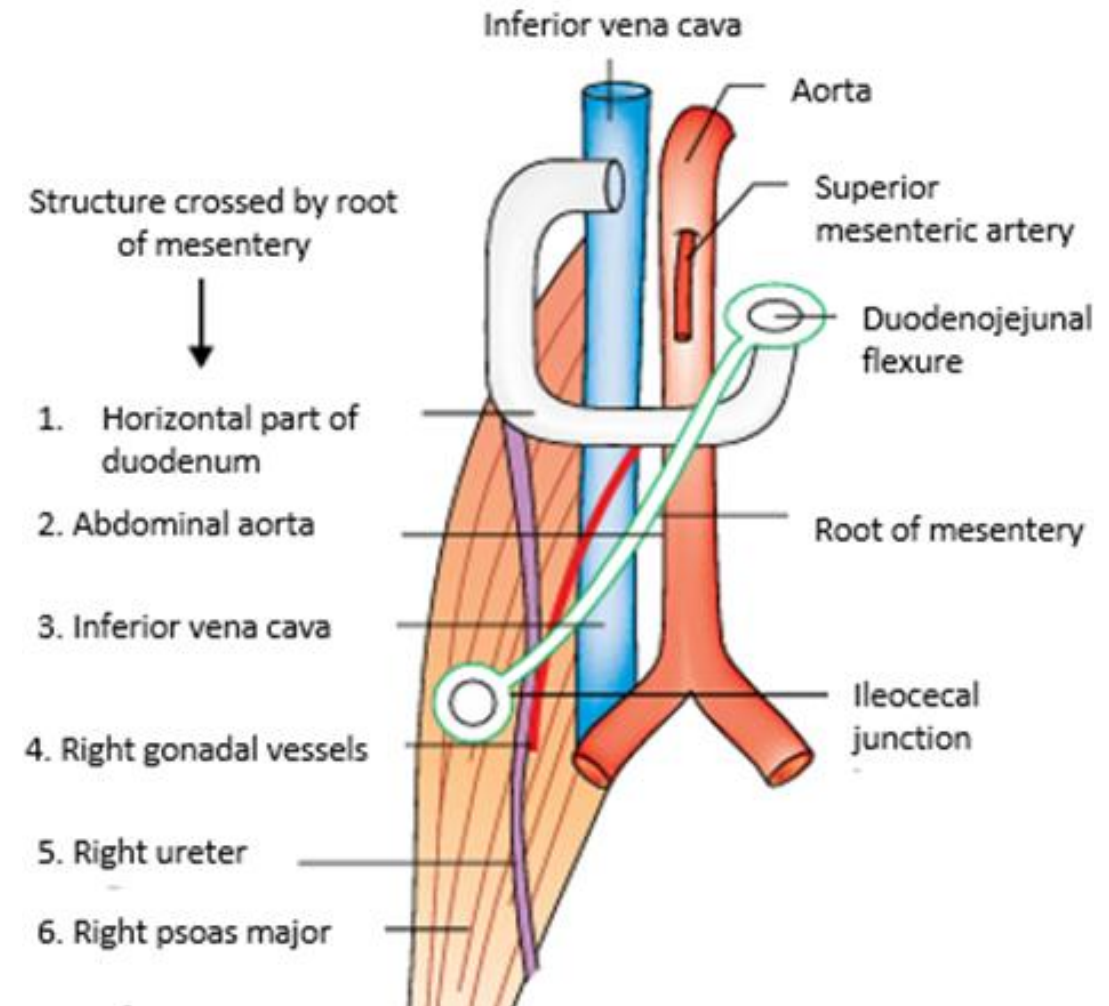
Mesentery (Mesentery of Small Intestine)

- ❑ It is a broad fan-shaped fold of the peritoneum that suspends the coils of the small intestine (jejunum and ileum) from the posterior abdominal wall. It has two borders: the attached border (or root) and long free edge (intestine border)
- ❑ the root divides the infracolic compartment into two parts: right and left
- ❑ **Free border (intestinal border):** It is about 6 m (20 feet) long and encloses the jejunum and ileum and accounts for the formation of folds (pleats) in it (a frill-like arrangement). Its length permits the free mobility of the loops of jejunum and ileum in the abdominal cavity.

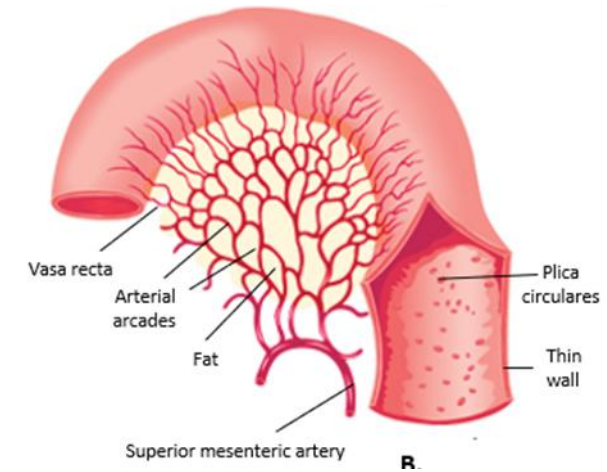
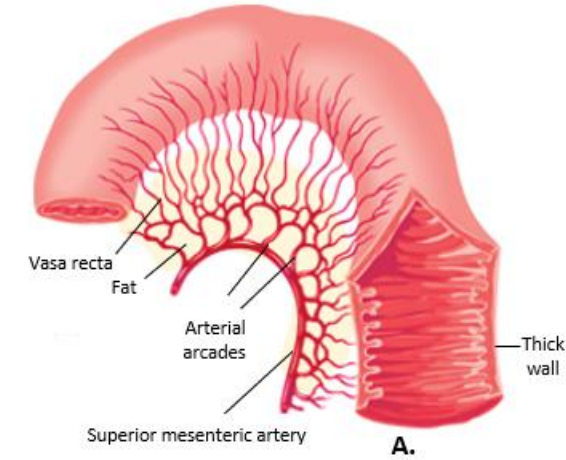


□ Root of the mesentery

- It is attached to an oblique line across the posterior abdominal wall, extending from the duodenojejunal flexure (left side of L2) to the ileocecal junction (right sacroiliac joint)
- It is crossed from above downward by the horizontal (third) part of the duodenum, abdominal aorta, inferior vena cava, right gonadal vessels, right ureter, and right psoas major muscle.
- The mesentery contains the superior mesenteric artery, vein, and lymphatic vessels, as well as nodes, fat, and nerve fibers.

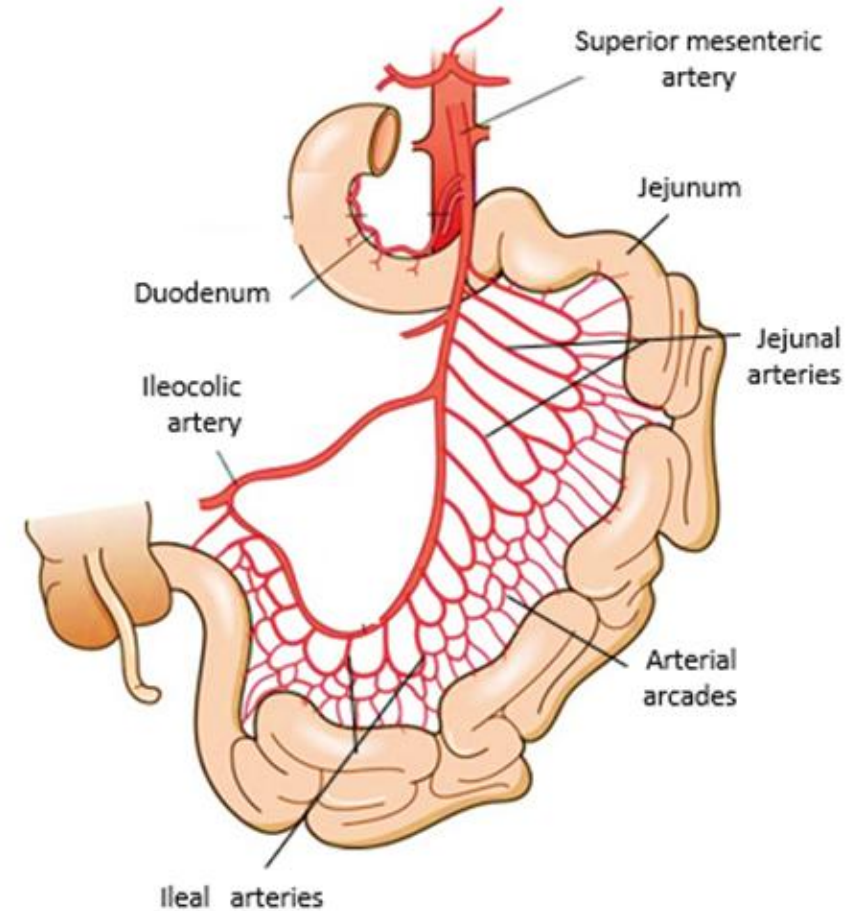


Feature	Jejunum	Ileum
Location	Left hypochondriac region and is the middle part of the small intestine	Right iliac region and the distal part of the small intestine
Length	About 2.5 meters	About 3.5 meters
Diameter	Wider	Narrower
Wall thickness	Thicker and more vascularized	Thinner and less vascularized
Color	Deeper red due to richer blood supply	Paler pink
Circular folds (Plicae)	Prominent and dense	Smaller and sparse
Lymphoid tissue	Few lymph nodes	Numerous Peyer's patches
Vascular supply	Richer blood supply Few arcades Long vasa recta	Less extensive blood supply Multiple arcades Short vasa recta
Fat in mesentery	Less fat near the intestinal wall, this allows clear spaces or "translucent windows called white windows	More fat, more fat reaching the wall, as a result, the white windows are absent



Arterial supply

- ❑ The jejunum and ileum are supplied by the jejunal and ileal branches (12–15 in number) of the superior mesenteric artery
- ❑ The terminal part of the ileum is supplied by the ileal branches of the ileocolic branch of the superior mesenteric artery.
- ❑ They divide into smaller branches that anastomose to form arterial arcades
- ❑ From the convexities of the terminal arcades, small parallel straight vessels called “vasa recta” arise and pass to the mesenteric border of the gut



Venous drainage

- ❑ The veins correspond to the branches of superior mesenteric artery and drain into the portal vein, which carries the products of protein and carbohydrates to the liver.

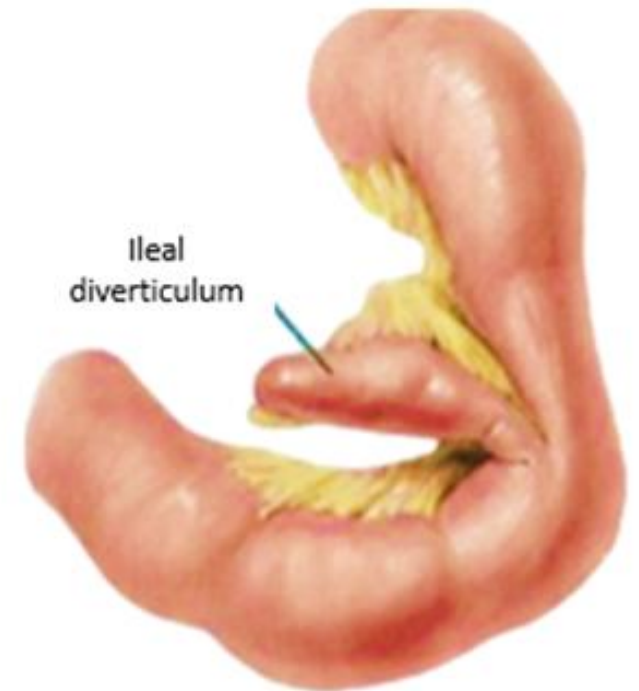
Lymphatic drainage

- ❑ The lymph vessels from the small intestine pass through a large number of mesenteric nodes (lymph nodes present in the mesentery) and finally drain into superior mesenteric nodes present around the origin of superior mesenteric artery.

CLINICAL CORRELATION

❖ **Meckel's diverticulum:** also known as ileal diverticulum, arises from the proximal part of the embryonic vitellointestinal duct.

- It can be remembered using the mnemonic "rules of two"
- It occurs in 2 percent of the subjects.
- It is usually 2 inches (5 cm) long.
- It is situated about 2 feet (60 cm) proximal to the ileocecal junction.
- It is more common in men.
- The caliber and structure of its wall are similar to that of the ileum.



Thank You